

Replication Package

Intermediate Input Prices and the Labor Share

Juanma Castro-Vincenzi Benny Kleinman

September 2026

Contents

1	Data Availability Statement	2
2	Contents of the Replication Package	9
3	Data Construction and Analysis Variables	10
3.1	U.S. industry data and derived variables	11
3.2	Regional analysis data (Table 2)	14
3.3	Cross-country analysis data (Table 4 and Figures F.1–F.2)	15
3.4	Disaster instrument data (Tables D.1–D.3)	16
3.5	Quantitative-model variables and numerical construction	16
4	Software Requirements	17
5	Instructions for Replicators	19
6	Where Output Is Saved, and How It Maps to the Paper	20
7	Data Citations	22

1 Data Availability Statement

All data used in this paper are secondary data obtained from public sources. No primary data were collected. All raw data files are included in the replication package. Whenever the raw data file is in .dta, .xls or .xlsx format a copy is provided in .csv format. Table 1 lists the raw data families; Table 3 lists the supporting concordances. Together they identify what each file contains, where it came from, and its access date where recorded. The five BEA workbooks used to construct the quantitative model inputs are included in `data/raw/model_inputs/`; their original access dates were not recorded. Their embedded source dates and the construction steps are described below. Rebuilding these inputs requires no data downloads.

Rights and permissions

We certify that we have legitimate access to, and permission to use, the data used in this paper. The authors' code is released under the BSD 3-Clause License. The documentation and the authors' original contributions to derived data and outputs are released under CC BY 4.0, only to the extent that the authors hold those rights. These grants do not relicense third-party material or override its source-data terms.

The EM-DAT archive and its format-only CSV copy are distributed under CC BY-NC-ND 4.0, as detailed below and in `LICENSE.txt`. Other source datasets retain their original providers' terms.

EM-DAT archived source, access and reuse

The package includes *The EM-DAT Emergency Events Database Archive*, version 2.0, archived on 30 April 2026 at UCLouvain Dataverse (Delforge, Wathelet, Below, Lanfredi Sofia, Tonnelier, van Loenhout, and Speybroek, 2024). The archive covers 1900–2024 and contains 26,946 records and 47 variables. Its permanent identifier is <https://doi.org/10.14428/DVN/IOLTPH>. The complete workbook `260430_emdat_archive.xlsx` is included without content changes, under the filename expected by the existing code: `data/raw/disaster_iv/disaster_iv_emdat.xlsx`.

Access. No registration, payment or application is required to download this archived file. The included copy was downloaded on 9 September 2026 (UTC). To obtain the same source, open the DOI, select version 2.0 and download `260430_emdat_archive.xlsx`, or use the fixed file link <https://dataverse.uclouvain.be/api/access/datafile/38893>. Save it at the package path above. Its SHA256 checksum is `21c210e479623144cef32c11c0a57c21094ea66f4ad0a0a248d8239f3a5e9847`. The data-column definitions and archived documentation are available with the same repository deposit. No download is needed when using the bundled copy.

Non-proprietary copy. A UTF-8 CSV copy of the entire EM-DAT Data worksheet is included at `data/raw_csv_copy/disaster_iv/disaster_iv_emdat__EM-DAT_Data.csv`. It preserves all records, column names and cell values; only the file format changes. The original archived workbook has no separate EM-DAT Info worksheet.

Reuse conditions. The provider releases this archive under Creative Commons Attribution–NonCommercial–NoDerivatives 4.0 International (CC BY-NC-ND 4.0): <https://creativecommons.org/licenses/by-nc-nd/4.0/>. Noncommercial sharing of the unadapted archive is permitted with attribution and the licence notice retained. A faithful change of file format is permitted. The licence does not permit sharing adapted material under its grant; the authors' BSD and CC BY licences do not replace these EM-DAT terms. The source attribution, format-conversion notice and licence link are recorded in `LICENSE.txt`.

Analysis sample and verification. The disaster-instrument code and Table D.1 use events with Start Year 1991–2016, Disaster Group Natural, excluding Epidemic, Infestation and

Animal incident. The 8,484 selected events and all seven fields read by these routines (Disaster Group, Disaster Type, ISO, Start Year, Total Deaths, Country and Disaster Subtype) match the earlier 27 February 2026 living-data extract. An isolated run with the archived workbook and unchanged code reproduced the contents of Tables D.1–D.3 exactly. The complete source workbooks differ in unused fields, other years and row order; the February living-data extract is not the archived file and is not included in this package. The applicable redistribution terms are those of the archived product, rather than those of the living-data portal.

Table 1: Raw data files

File(s)	Description	URL	Accessed
aggregate/WPSID61.csv	Producer price index for processed goods for intermediate demand.	https://fred.stlouisfed.org/series/WPSID61	Mar 2026
aggregate/ 14 further FRED series	Producer, consumer and GDP price indices, exchange rates, manufacturing earnings and BLS productivity: WPSID62, WPSFD49201, WPSFD49207, PRS85006173, GDP, GDPDEF, DGDSRG3M086SBEA, LCEAMN01USQ189S, NAGIGP01EZQ661S, JPNGDPDEFQISMEI, CCUSMA02EZQ618N, CCUSMA02JPQ618N, GOODS0EU28M086NEST, PITGCG02EZA661N.	<a href="https://fred.stlouisfed.org/series/<ID>">https://fred.stlouisfed.org/series/<ID>	Mar 2026
aggregate/nipaQuarterly.csv	BEA National Income and Product Accounts, quarterly.	https://apps.bea.gov/national/Release/TXT/NipaDataQ.txt	Mar 2026
aggregate/fernald.xlsx	Fernald utilisation-adjusted quarterly TFP.	https://www.frbsf.org/economic-research/files/quarterly_tfp.xlsx	Mar 2026
aggregate/BEA_GDPbyIndustry_SIC_1947–1997.xls	BEA historical industry accounts: value added, gross output, compensation and wages by industry, 1947–1997.	https://apps.bea.gov/industry/xls/GDPbyInd_VA_SIC.xls	Jul 2026
aggregate/BEA_GDPbyIndustry_ValueAdded.xlsx	BEA GDP by Industry, value added by industry, 1997–2018.	current release: https://www.bea.gov/data/gdp/gdp-industry	Oct 2019
aggregate/BEA_GDPbyIndustry_GrossOutput.xlsx	BEA GDP by Industry, gross output by industry, 1997–2018.	current release: https://www.bea.gov/data/gdp/gdp-industry	Oct 2019

Table 1 continued

File(s)	Description	URL	Accessed
aggregate/BEA_NIPA_Section6_Dec2019.xlsx	NIPA Section 6, December 2019 workbook; the annual Tables 6.2D and 6.3D used here were published July 30, 2019.	current release: https://apps.bea.gov/national/Release/XLS/Survey/Section6all_xls.xlsx	Not verified
model_inputs/income_emp_tab.xlsx	BEA NIPA Section 6, archived workbook used for model compensation and wages.	https://apps.bea.gov/national/Release/XLS/Survey/Section6All_xls.xlsx	Not recorded
model_inputs/gross_output_tab.xlsx	BEA GDP by Industry, archived gross output levels and quantity indexes for the model.	https://apps.bea.gov/industry/Release/XLS/GDPxInd/GrossOutput.xlsx	Not recorded
model_inputs/int_input_tab.xlsx	BEA GDP by Industry, archived intermediate input levels and price indexes for the model.	https://apps.bea.gov/industry/Release/XLS/GDPxInd/IntermediateInputs.xlsx	Not recorded
model_inputs/value_added_tab.xlsx	BEA GDP by Industry, archived value added levels for the model.	https://apps.bea.gov/industry/Release/XLS/GDPxInd/ValueAdded.xlsx	Not recorded
model_inputs/fixed_assets_tab.xlsx	BEA Fixed Assets Section 3, archived capital, depreciation and investment data for the model.	https://apps.bea.gov/national/FixedAssets/Release/XLS/Section3All_xls.xlsx	Not recorded
external_gas_prices.csv	EIA annual natural gas import prices for HS 271111 and 271121.	https://www.eia.gov/dnav/ng/hist/n9103us3a.htm	Mar 2026
BEA-KLEMS/BEA-BLS-industry-level-production-account-1987-2021.xlsx	BEA-BLS Integrated Industry-Level Production Account, the U.S. KLEMS accounts.	https://www.bea.gov/sites/default/files/2023-09/BEA-BLS-industry-level-production-account-1987-2021.xlsx	Sep 2023
BIS/mod.csv	Daily US dollar exchange rates.	https://data.bis.org/topics/XRU	Oct 2023
census/NAICSUseDetail.txt	BEA 1997 benchmark Input-Output Use table at NAICS detail.	https://apps.bea.gov/industry/zip/ndn0306.zip	Jul 2026

Table 1 continued

File(s)	Description	URL	Accessed
census/concentration_ratios/1997/, census/concentration_ratios/2002/, census/concentration_ratios/2007/, census/concentration_ratios/2012/ — 46 .dat files	Economic Census concentration ratios (CR4, CR8, CR20, CR50, HHI).	URLs listed in census/concentration_ratios/SOURCE_URLS.txt	Jul 2026
cepii_tuv/tuv_96_x_2000.csv through tuv_96_x_2016.csv (17 files)	CEPII Trade Unit Values, HS96 export regime	https://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=2	Jul 2026
comtrade/comtrade_1991.csv through comtrade_2016.csv (6 files)	UN Comtrade U.S. imports from world, HS 1992 (H0), covering 1991–2020 in six five-year blocks. The 182,348 rows include HS6 records and other aggregation levels.	https://comtradeplus.un.org/	Feb 2026
disaster_iv/disaster_iv_baci_trade_1997.csv, disaster_iv_baci_country_codes.csv	CEPII BACI bilateral trade for 1997 and its country dictionary - version V202501.	https://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=37	Mar 2026
disaster_iv/disaster_iv_emdat.xlsx	Official EM-DAT Archive, version 2.0, 30 April 2026; 1900–2024. CC BY-NC-ND 4.0.	https://doi.org/10.14428/DVN/IOLTPH	9 Sep 2026 (UTC)
EUKLEMS/EUKLEMS_2025_vintage_national_accounts.dta, ..._capital_accounts.dta	EU KLEMS 2025 release.	https://euklems-intanprod-llee.luiss.it/	Mar 2026
EUKLEMS/EUKLEMS_2020_vintage.dta	EU KLEMS 2019 release.	https://euklems.eu/bulk/Statistical-National-Accounts.dta	Mar 2026
EUKLEMS/WorldKLEMS_Canada_2012.xlsx, WorldKLEMS_Korea_2015.xlsx	World KLEMS accounts for Canada (2012 release) and Korea (2015 release).	http://www.worldklems.net/data/basic/	Mar 2026
EUKLEMS/WIOD/ 43 workbooks, AUS_NIOT_nov16.xlsx through USA_NIOT_nov16.xlsx	WIOD 2016 release national input-output tables.	https://www.rug.nl/ggdc/valuechain/wiod/wiod-2016-release	Mar 2026

Table 1 continued

File(s)	Description	URL	Accessed
misc/CMO-Historical-D ata-Annual.xlsx	World Bank Pink Sheet annual commodity prices - “Crude oil, average” in nominal and constant 2010 US\$.	https://www.worldbank.org/en/research/commodity-markets	Jul 2026
NBER-CES/nberces_1958-2018.dta, nberces-1958_2018_naics2012.dta	NBER-CES Manufacturing Industry Database, NAICS-1997 and NAICS-2012 versions, release 2021a.	https://www.nber.org/research/data/nber-ces-manufacturing-industry-database	Mar 2021
NBER-CES/nberces_1958-2011_naics.dta	NBER-CES Manufacturing Industry Database.	https://data.nber.org/nberces/nberces5811/naics5811.dta	Jul 2026
regional_data/QCEW/1990.annual.singlefile.csv through 2020.annual.singlefile.csv (31 files)	BLS Quarterly Census of Employment and Wages annual averages by area, ownership and industry; county records are aggregated to commuting zones by the cleaning code.	https://data.bls.gov/qcew/data/files/	Jul 2026
regional_data/bea_county/CAGDP2__ALL_AREAS_2001_2020.csv, CAINC30__ALL_AREAS_1969_2020.csv	BEA county GDP by industry and county economic profile.	current release: https://www.bea.gov/data/gdp/gdp-by-county and https://www.bea.gov/data/income-saving/personal-income-by-county	Jun 2022
The county files bundled here are the historical extracts used by the code. The linked BEA pages provide current releases and access to previously published estimates; current downloads may differ from the bundled vintage. An exact historical download URL for these two extracts was not recorded.			
trade/imp_detl_yearly_89n.dta through 118n.dta and the exp_detl_yearly_equivalents (60 files)	Schott U.S. import and export data by detailed product, 1989–2018.	https://sompks4.github.io/sub_data.html	Dec 2019

Inputs to the quantitative model

The Stata script `do/data_cleaning/model_inputs.do` constructs the two files read by Julia from the five archived BEA workbooks in `data/raw/model_inputs/`: `quant_model/data/data_manufacturing.csv`, covering total manufacturing, and `quant_model/data/subsector_data.csv`, covering the same variables for 19 manufacturing subsectors plus the durable, nondurable and total aggregates. The files contain annual observations for 1998–2023: 26 rows for total manufacturing and 572 rows for the 22-category panel. Prebuilt copies are included; the Stata driver rebuilds them before the Julia model is run.

The source workbooks are the 2025 vintages of BEA’s NIPA Section 6, GDP by Industry and

Fixed Assets Section 3 (Bureau of Economic Analysis, 2025c,b,a). The three GDP by Industry workbooks identify September 25, 2025 as their publication date. The NIPA workbook identifies September 26, 2025 as its publication date and December 22, 2025 as its file-creation date. The Fixed Assets workbook records a file-creation date of September 15, 2025; this is not a recorded publication or access date. The original download dates are unknown. The URLs in Table 1 serve changing BEA releases; use the included workbooks to reproduce the supplied inputs at their exact source vintage.

Table 2 identifies the 12 worksheets used in construction. The script matches the 22 manufacturing categories by their industry names, using the first occurrence in the manufacturing section when an aggregate name is repeated elsewhere in a BEA worksheet. It reshapes the annual series, retains 1998–2023, and merges the series by year and industry. `Dep` is current-cost depreciation divided by the same year’s current-cost net capital stock. The script retains the supplied inputs’ sector encoding and Stata float storage precision and rounds numeric variables to 0.001 before export. It checks the merge matches, nonmissing values and the 572-row panel and 26-row manufacturing series. Intermediate files are written to `data/temp/model_inputs/`.

Plain-text copies of all 110 worksheets in the five workbooks, including their metadata worksheets, are provided in `data/raw_csv_copy/model_inputs/`. Each filename consists of the workbook stem, two underscores and the worksheet name; for example, `gross_output_tab__TG0105-A.csv`. The builder reads the archived Excel workbooks directly and needs no internet access or user-written Stata commands.

Table 2: Sources for the quantitative model inputs

Variable	Description	BEA source
Comp	Compensation of employees, manufacturing, millions of current dollars.	NIPA Table 6.2D; <code>income_emp_tab.xlsx</code> , sheet T60200D-A.
Wage	Wages and salaries per full-time equivalent employee, dollars per employee per year.	NIPA Table 6.6D; <code>income_emp_tab.xlsx</code> , sheet T60600D-A.
VA	Value added, manufacturing, millions of current dollars.	GDP by Industry, Value Added by Industry; <code>value_added_tab.xlsx</code> , sheet TVA105-A.
Mat	Intermediate inputs, manufacturing, millions of current dollars.	GDP by Industry, Intermediate Inputs by Industry; <code>int_input_tab.xlsx</code> , sheet TII105-A.
G0mil	Gross output, manufacturing, millions of current dollars.	GDP by Industry, Gross Output by Industry; <code>gross_output_tab.xlsx</code> , sheet TG0105-A.
Capitalmil	Current-cost net stock of private fixed assets, manufacturing, millions of dollars at year end.	Fixed Assets, Table 3.1ESI, Current-Cost Net Stock of Private Fixed Assets by Industry; <code>fixed_assets_tab.xlsx</code> , sheet FAAt301ESI-A.
Investmentmil	Investment in private fixed assets, manufacturing, millions of current dollars.	Fixed Assets, Table 3.7ESI, Investment in Private Fixed Assets by Industry; <code>fixed_assets_tab.xlsx</code> , sheet FAAt307ESI-A.

Table 2 continued

Variable	Description	BEA source
Dep	Annual depreciation divided by current-cost net stock; a fraction, not a percentage.	Fixed Assets, Table 3.4ESI (Current-Cost Depreciation) divided by Table 3.1ESI; <code>fixed_assets_tab.xlsx</code> , sheets FAAt304ESI-A and FAAt301ESI-A.
GO_ind	Gross-output chain-type quantity index, 2017=100.	GDP by Industry, Chain-Type Quantity Indexes for Gross Output by Industry; <code>gross_output_tab.xlsx</code> , sheet TG0103-A.
Pm	Intermediate-input chain-type price index, 2017=100.	GDP by Industry, Chain-Type Price Indexes for Intermediate Inputs by Industry; <code>int_input_tab.xlsx</code> , sheet TII104-A.
Cap_ind	Net-capital-stock chain-type quantity index, 2017=100.	Fixed Assets, Table 3.2ESI, Chain-Type Quantity Indexes for Net Stock of Private Fixed Assets by Industry; <code>fixed_assets_tab.xlsx</code> , sheet FAAt302ESI-A.
Inv_ind	Investment chain-type quantity index, 2017=100.	Fixed Assets, Table 3.8ESI, Chain-Type Quantity Indexes for Investment in Private Fixed Assets by Industry; <code>fixed_assets_tab.xlsx</code> , sheet FAAt308ESI-A.

Concordances and crosswalks

Table 3 lists the mapping files used to link classifications. All are included in the package. In addition to code correspondences, the supplied files retain population metadata (`cz00pop`) and establishment, shipment, payroll and employment mapping weights.

Table 3: Concordances and crosswalks

File	Original source	Notes
<code>Concordances/naics_97to12.dta</code> , <code>naics_02to07.dta</code> , <code>naics_07to12.dta</code>	U.S. Census Bureau	NAICS classification concordances: 1997–2012, 2002–2007 and 2007–2012.
<code>Concordances/conc_naics97_naics12.dta</code>	Becker, Gray, and Marvakov (2021)	NAICS 1997 to 2012
<code>Concordances/naics2012_to_bea.dta</code>	Bureau of Economic Analysis	NAICS 2012 to BEA sector, summary and detail industry codes; Table C.7 uses <code>bea_summary</code> .
<code>Concordances/IOT_NAICS.dta</code>	Bureau of Economic Analysis	I-O industry to 6-digit NAICS.
<code>Concordances/HS696_to_IOT.dta</code>	Bureau of Economic Analysis	HS 1996 commodities to I-O commodities.

Table 3 continued

File	Original source	Notes
Concordances/HS696_Commodities.dta	Fally and Sayre (2022)	Commodity list with HS 1992 and HS 1996 codes.
Concordances/cty_czone.dta	Dorn (2009)	County FIPS to 1990 commuting zone.
EUKLEMS/WIOD/JobID-19_Concordance_H1_to_I3.CSV	World Integrated Trade Solution (WITS)	HS 1996 to ISIC Revision 3.
misc/naicsnames12.dta, naics3names12.dta	U.S. Census Bureau	NAICS 2012 industry names.

2 Contents of the Replication Package

Path	Contents
README.pdf	This document
LICENSE.txt	Terms of use for the code and data in this package
README.tex, data_citations.bib, ecta-fullname.bst	L ^A T _E X source of this document
do/main.do	Stata driver: constructs model inputs and runs all empirical analyses
do/setup.do	Validates the current package root and initializes Stata paths
do/data_cleaning/	21 scripts: raw data → model inputs and analysis data; <code>model_inputs.do</code> runs first
do/figures/	13 Stata analysis scripts, one per figure (or figure family), the shared style helper <code>graph_style.do</code> , and an editable schematic drawing, <code>fig2.tex</code>
do/tables/	19 scripts, one per table
data/raw/	Source files, including preserved downloads and assembled source extracts
data/raw/model_inputs/	Five archived BEA workbooks used to construct the quantitative model inputs
data/raw_csv_copy/	Plain-text <code>.csv</code> copies of every raw file supplied in <code>.dta</code> , <code>.xls</code> or <code>.xlsx</code> format. Provided for readability only; no program reads from this folder
data/dta/	Analysis data built by the cleaning scripts from <code>data/raw/</code> . Created by <code>do/main.do</code> , which rebuilds its contents
data/temp/	Scratch space created by <code>do/main.do</code> when needed. The model-input builder creates its intermediates in <code>model_inputs/</code>
output/tables/	19 generated <code>.tex</code> tables and the supporting Table D.1 country-list CSV
output/figures/	25 generated <code>.png</code> panels and the supplied Figure 2 schematic

Path	Contents
<code>quant_model/</code>	Julia code for the quantitative model
<code>quant_model/data/</code>	Two prebuilt model-input CSV files, regenerated from the archived BEA workbooks by <code>do/data_cleaning/model_inputs.do</code>
<code>quant_model/Project.toml</code> , <code>Manifest.toml</code>	Pinned Julia environment used by the supplied model and the verified replication run
<code>quant_model/results/</code>	Model figures and table

Initial package contents. Generated analysis data, figures, tables, model results and execution logs are omitted from the submission archive; the documented drivers recreate them. The output directories are retained, including the initially empty `quant_model/results/` directory required by the Julia driver. The supplied Figure 2 schematic, `output/figures/fig2.png`, is retained because it does not require computational reproduction. The two prebuilt model-input CSV files in `quant_model/data/` are also retained as documented below.

The raw data directories are: `aggregate/` (FRED, NIPA, Fernald), `BEA-KLEMS/`, `BIS/`, `census/`, `cepii_tuv/`, `comtrade/`, `Concordances/`, `disaster_iv/`, `EUKLEMS/` (incl. WIOD and World KLEMS), `misc/`, `model_inputs/` (five archived BEA model workbooks), `NBER-CES/`, `regional_data/` (QCEW and BEA county), and `trade/`.

The Stata-derived CSV copies preserve the stored numeric values, using up to 17 significant digits instead of the variables' display formats. Numeric value labels are retained as their underlying codes; dates retain their numeric Stata representation. String values, including leading zeros, are preserved. Standard numeric missing values are empty fields; extended missing values retain their literal codes `.a` through `.z`. When importing these CSVs, keep the source string columns as strings and read double-precision numeric columns as doubles. Source storage types, display formats, variable and value labels, checksums, and full-value validation results are recorded in `data/raw_csv_copy/stata_csv_validation.json`.

The optional conversion utility `do/utilities/export_stata_csv.py` checks all raw `.dta` files and regenerates any CSV that fails exact comparison at the source storage precision. It was tested with Python 3.12, pandas 3.0.5 and NumPy 2.5.3. These dependencies are needed only to rerun this conversion utility; the Stata and Julia analysis drivers do not use it. From the replication root, run:

```
python do/utilities/export_stata_csv.py --source data/raw --output data/raw_csv_copy
```

Add `--verify-only` to validate the supplied copies without replacing them. The utility records its runtime and validates row and column order, every string and missing code, and every numeric value against the raw inputs.

3 Data Construction and Analysis Variables

This section links the source files above to the datasets and variables used by the analysis. All paths are relative to the package root. The listed scripts are included under `do/data_cleaning/`; `do/main.do` runs them in their required order. Source data remain in `data/raw/`; transformations are written to `data/dta/`, `data/temp/` or `quant_model/data/`.

The definitions below supplement the source labels, workbook units and provider documentation. Classification codes are identifiers. Years are calendar years unless another frequency is

specified; logarithms are natural logs. Shares and ratios are fractions unless a percentage or index base is specified. Missing values are not zero unless explicitly replaced during construction.

3.1 U.S. industry data and derived variables

Main industry panel and units. `do/data_cleaning/create_ss_dataset.do` builds `data/dta/mainregfile.dta`, indexed by six-digit NAICS-2012 industry and year. Its NBER-CES source uses millions of dollars for payroll, production-worker wages, shipments, materials, value added and capital; capital is a real stock. Total and production-worker employment are in thousands of workers, and production-worker hours are in millions of hours. The source price deflators `pimat`, `piship`, `piinv` and `pien` have base 2012= 1.

Analysis variable	Construction and meaning
<code>labshr</code> , <code>llabshr</code>	<code>pay/vadd</code> , and its log: payroll share of value added.
<code>mat2va</code> , <code>mat2lab</code>	<code>matcost/vadd</code> and <code>matcost/pay</code> ; the latter's log is <code>lmat2lab</code> .
<code>w</code> , <code>lw</code>	<code>prodw/prodh</code> , and its log: production-worker wages per hour (dollars per hour).
<code>cap2lab</code> , <code>lcap2lab</code>	<code>cap/prodh</code> , and its log: real capital stock per production-worker hour. This control has a different denominator from Table C.3's <code>1K2L=log(cap/emp)</code> , which uses total employees.
<code>prodshr</code>	<code>prode/emp</code> : production workers divided by total employment.
<code>lpimat</code> , <code>lpiinv</code>	Logs of the NBER-CES materials and investment price deflators.
<code>impen</code> , <code>impen_china</code>	<code>im/(vship+im-ex)</code> and <code>im_china/(vship+im-ex)</code> : imports divided by domestic absorption; the China measure changes the numerator only. All four monetary inputs use millions of dollars.
<code>vadd_90</code> , <code>mat2va_90</code>	Industry value added and materials/value added in 1990, held fixed within industry. The suffixes <code>_58</code> , <code>_70</code> , <code>_72</code> , <code>_80</code> and <code>_97</code> identify analogous reference-year values used in other exercises.
<code>treat</code>	<code>mat2va_90*lpimat</code> : the main endogenous regressor.
<code>naics3</code> , <code>n3y</code>	The first three NAICS digits, and the three-digit-industry/year group identifier used for subsector-by-year fixed effects.

Table 1 uses years 1991–2016 and a nonmissing `shift_share_ct`; regressions additionally require nonmissing outcomes and regressors, plus positive weights in weighted columns. Those columns use `vadd_90`. The individual table scripts give the controls, fixed effects and sample restrictions for each column. Tables C.3 and C.5 create additional logged ratios locally: materials/value added, value added/sales, materials/payroll, materials/real capital, real capital/value added, and real capital/total employees, as applicable to each table.

Schott trade inputs. `do/data_cleaning/trade.do` reads the 1989–2018 import and export files under `data/raw/trade/`, retaining records whose supplied NAICS code begins with 3. It sums general import value (`gen_val_yr`) and total export value (`all_val_yr`) by industry and year and divides by 10^6 . China imports use country code 5700; the additional `im_nafta` field uses Canada (1220) and Mexico (2010). The resulting files are `data/dta/imports_1989–2018_naics.dta` and `data/dta/exports_1989–2018_naics.dta`.

The main-panel builder links the NAICS-1997 NBER-CES panel and these external series to NAICS-2012 through `data/raw/Concordances/conc_naics97_naics12.dta`, retaining links with `pay9712>0.7`. When multiple source industries map to one NAICS-2012 industry, import and export values are raw sums; exposure shares and shift-share variables are means weighted

by source-industry shipments in the corresponding year. The resulting industry/year records are merged with the NAICS-2012 NBER-CES outcomes and the concentration data.

Commodity prices and cleaning. `do/data_cleaning/cep_cleaned.do` reads the six bundled Comtrade files (U.S. imports from the world, HS92, 1991–2020). It retains six-digit leaf records with code greater than 10000, positive reported quantity and trade value, usable unit values, denominator quantity greater than 1 and trade value at least US\$1,000. Its exact lists of product-specific repairs are declared near the beginning of that script.

Unit values start as trade value divided by reported quantity. For the listed quantity-scale cases, 2000–2005 quantities are multiplied by 1000. For crude oil and the two gas codes, positive net weight in kilograms is used when present. LNG (271111) and gaseous natural gas (271121) use the annual paths in `data/raw/external_gas_prices.csv` when available: EIA series `n9103us3A` and `n9100us3A`, respectively, both reported in dollars per thousand cubic feet. Their source pages are <https://www.eia.gov/dnav/ng/hist/n9103us3a.htm> and <https://www.eia.gov/dnav/ng/hist/n9100us3a.htm>. Subsequent normalization uses within-product price changes, not a common physical unit across products.

For non-energy products, the quantity-unit code must match its 1997 value. A consecutive-year unit-value ratio is discarded if it lies below 0.2 or above 5 times that product’s median consecutive-year ratio. Energy products (HS chapter 27) are exempt from those unit-consistency and growth screens. For the designated residual series, the code computes annual product-level medians of CEPII export unit values from the 2000–2016 files, then sets the post-1999 path to the observed 1999 unit value times the CEPII median in year t divided by its 2000 median. Thus the spliced 2000 level equals the 1999 anchor. Replacement requires the relevant observed anchor and CEPII values.

Each surviving product price is normalized to 1997 = 100; observations without that index are dropped. After mapping to the commodity list, non-energy product series with any index value above 20,000 are removed. HS92-to-HS96 links use `data/raw/Concordances/HS696_Commodities.dta`; HS96-to-IO links and the supplied within-commodity `weight` variable use `data/raw/Concordances/HS696_to_IOT.dta`. The cleaner saves the mapped product data as `data/dta/Commodity_Prices_HS96.dta` and the weighted-mean IO-commodity/year price indexes as `data/dta/Commodity_Prices_Weight_comtrade_new.dta`. The full cleaned HS92 data and the intermediate filter-count file are also written by this script.

Input-output exposures and instruments. `do/data_cleaning/create_shares.do` reads `data/raw/census/NAICSUseDetail.txt`. Compensation is the V00100 row; after excluding final-demand and special-category rows/columns, missing industry/commodity combinations are filled with zero. Direct exposure is `use_value` divided by the using industry’s summed intermediate use plus compensation. The file retains the supplied use values without subtracting their included margins. The alternative exposure matrix is $A(I - A)^{-1}$, using the cost-share matrix A ; both exposures are saved in `data/dta/IOLinkages.dta`.

`do/data_cleaning/create_ss_instrument.do` joins these exposures to available commodity price indexes P , multiplies each exposure by $\log(P/100)$, and sums by industry and year. `shift_t_share_ct` uses direct exposures; `shift_share_leontieff_ct` uses the alternative matrix. The `shift_share_sums1_ct` variant divides use values by total use across commodities present in the joined data in 2000 and carries those weights across years. The direct-exposure variants `shift_share_e_ct` and `shift_share_ne_ct` sum energy and non-energy components separately. `share_ct` sums direct shares over the commodity records available in each year. `data/raw/Conco`

`rdances/IOT_NAICS.dta` maps these results to NAICS-1997 in `data/dta/shiftshare_ct.dta`; the main-panel mapping described above then produces the NAICS-2012 variables used in regressions.

Industry concentration. `do/data_cleaning/build_censuscr_raw.do` extracts the Census 1997, 2002, 2007 and 2012 concentration files, retains six-character NAICS industries and the all-establishments tax-status universe where specified, and converts flagged values to missing. It writes `data/dta/censuscr_97-02-07-12_n6.dta`. `do/data_cleaning/concentration.do` retains manufacturing and maps 1997 directly to 2012, 2002 through 2007 to 2012, and 2007 to 2012 using the supplied Census concordances. For multiple source industries mapped to one destination, company counts and sales are summed; positive, nonmissing concentration ratios and HHI values are combined as unweighted harmonic means. The output is `data/dta/concentration.dta`. The main-panel builder divides CR4, CR8, CR20 and CR50 by 100 to express sales shares as fractions. HHI-50 retains the Census index scale; `lhh50` is its natural log. Table C.10 uses CR4 and log HHI-50 on the common nonmissing sample in the four Census years.

BEA-KLEMS and the BEA-subsector specification. `do/data_cleaning/BEA_KLEMS.do` reshapes worksheets 3–30 of the bundled 1987–2021 production-account workbook, harmonizes industry names, merges its NAICS-code sheet and excludes state and local government. It saves `data/dta/BEA-BLS_klems_clean.dta`. Nominal compensation, value added and output are in millions of dollars; the quantity series, including labor hours, are indexes with 2012= 100.

Figures 3, 4 and C.2 use total college plus non-college labor compensation for labor costs and materials plus energy compensation for materials. Implicit prices divide the corresponding nominal expenditure by its quantity index. Materials and energy price growth is combined using the average of current and lagged expenditure shares (Tornqvist weights), with analogous expenditure-share aggregation across industries. Labor shares are total labor compensation divided by value added.

`do/tables/tabC7.do` separately maps the main panel to BEA summary industries. It applies value-added-weighted sums to NBER-CES quantities and value-added-weighted means to prices and instruments before merging BEA outcomes. Missing aggregated import/export values are set to zero in this specification. Its labor share uses BEA compensation/value added, whereas baseline materials intensity uses the aggregated NBER-CES `matcost/vadd`. Its materials price is BEA materials compensation/materials quantity, without adding energy. The services share is services compensation divided by services plus materials plus energy compensation; the services price divides services compensation by its quantity index. Wage and capital controls divide compensation and the aggregated NBER-CES capital stock, respectively, by the BEA labor-hours index. Reference-year intensity and weighted regression weights use 1990.

Oil instrument. `do/data_cleaning/oilprices.do` reads the nominal and real annual “Crude oil, average” columns from the bundled World Bank Pink Sheet, retains years through 2021 and writes `data/dta/oilprices.dta`. Real oil prices are in constant 2010 US dollars per barrel. Table 3 uses 1970–1980: its treatment is 1972 materials/value added times log materials price, and its instrument is 1972 energy expenditure/value added times log real oil price. Weighted columns use 1972 value added.

Aggregate descriptive series. `do/data_cleaning/agg_labour_shares.do` constructs four quarterly shares: NIPA compensation divided by compensation plus rental income, corporate profits, net interest and depreciation; nonfinancial-corporate compensation/gross value added; one minus Fernald’s capital-share series; and the BLS nonfarm-business index rescaled to 0.639 in 2000Q1.

Their unweighted arithmetic mean is `ls_mean` in `data/dta/agglabshr.dta`. `do/data_cleaning/agglabshr.dta` averages the listed FRED series to quarters in `data/temp/fred_prices.dta`. Figure 1 then averages quarterly series to years, reads the bundled annual WPSID61 series, divides processed and unprocessed intermediate-input PPIs by the PCE-goods price index, and normalizes both relative prices to 2000=100.

For Figure B.1, `do/data_cleaning/bea_historical.dta` assembles `data/dta/bea_historical.dta`: BEA SIC-basis accounts supply years through 1997 and modern BEA accounts supply 1998–2018; the 1972-SIC source wins at the overlapping 1987 observation. NBER-CES totals use the NAICS-1997 file before 2012, the NAICS-2012 file from 2012, and the separate 1958–2011 release for 2007–2011; NBER-CES values after 2016 are set to missing for this comparison. BEA intermediate expenditure in Figure B.1 is gross output minus value added.

3.2 Regional analysis data (Table 2)

`do/data_cleaning/qcew_panels.dta` reads the 1990–2020 annual QCEW CSV files in `data/raw/regional_data/QCEW/`. It retains private ownership (`own_code=5`), county records at sector and four-digit NAICS levels (`agglvl_code=74,76`), excludes area codes containing 999 and zero-employment records, and maps county FIPS to 1990 commuting zones with `data/raw/Concordances/cty_czone.dta`. It sums employment, establishments and annual payroll, and averages weekly wages using employment weights. The outputs are `data/dta/QCEW_naics2_panel.dta` and `data/dta/QCEW_naics4_panel.dta`, indexed by commuting zone, industry code and year. Payroll is in dollars; weekly wages are dollars per week.

`do/data_cleaning/bea_county.dta` reads the CAGDP2 and CAINC30 CSV files in `data/raw/regional_data/bea_county/`, retains county FIPS, reshapes annual values through 2020, and converts nonnumeric value markers to missing. Its outputs are `data/dta/bea_county_gdp.dta` and `data/dta/bea_county_otheraggdata.dta`, indexed by county `gcode` and year. CAGDP2 GDP values are in thousands of dollars.

`do/data_cleaning/regional.dta` combines these panels with `data/dta/mainregfile.dta`. It first averages six-digit industry log price changes and commodity instruments to four-digit NAICS using 2001 materials-cost weights. Changes are measured from each industry’s first available observation from 2001 onward. It then averages four-digit prices and instruments within each commuting-zone/year using that year’s industry payroll weights. National industry materials costs, shipments and trade are allocated to regions using their current-year shares of industry payroll. County codes are harmonized using the recodes in the script; matched county data are summed to commuting zones. The merged panel retains commuting zones with all 18 years from 2001–2018.

The output `data/dta/regional.dta` has key `cz1990`, `year`. `LS` is manufacturing payroll divided by manufacturing GDP, after converting payroll to thousands of dollars; `manuf_share` is manufacturing employment divided by total private employment. `impen` and `impen_china` divide the corresponding allocated imports by shipments plus imports minus exports. `M2VA` is allocated materials costs divided by manufacturing GDP (materials costs are converted from millions to thousands of dollars). `treat` and `iv` multiply the regional price change `lpm` and instrument change `lpmiv`, respectively, by `M2VA2001`. The regression weight `ww` is 2001 manufacturing GDP, distinct from the annual payroll weights inside the indices. `do/tables/tab2.dta` uses log labor share, log weekly wages, `lpk`, and the additional employment/trade controls specified there; its usable estimation observations cover 2001–2016. It writes `output/tables/tab2.tex`.

3.3 Cross-country analysis data (Table 4 and Figures F.1–F.2)

`do/data_cleaning/crosscountry_panel_build.do` combines the EU-KLEMS national and capital accounts in `data/raw/EUKLEMS/`, the historical file `EUKLEMS_2020vintage.dta`, and the Canada and Korea World KLEMS workbooks. Country, industry and year identify records. Missing or zero EU-KLEMS price indices are filled using the historical vintage scaled by the mean overlap ratio for that country/industry; series with at most three usable newer-vintage price observations use the historical series directly where available. Missing compensation and hours are filled directly. Canadian and Korean sectors are mapped to the common manufacturing groups by summing levels and using current-year value-added weights for aggregated price indices. Korean price indices are nominal/real ratios normalized to 2000=100. Canada has no separate C21 or C27 observations: those activities remain within its combined C20 and C26 source categories. The constructed panel sets investment and capital-account fields to missing for Canada and Korea.

The intermediate `data/dta/klems_panel_2025.dta` retains source currency units. It defines `labshr=comp/va`, `mat2va=ii/va`, `w=comp/h.empe` (compensation per employee hour in source units), and `pmat2pva=ii_pi/va_pi`; the variables `llabshr`, `lmat2va`, `lw`, `lii_pi`, `lva_pi`, `lgo_pi` and `lpmat2pva` are natural logs of the corresponding levels. Nonpositive value added, intermediate inputs, compensation or employment, and compensation above value added, are excluded during construction.

`do/data_cleaning/exchange_rates.do` averages the nonmissing daily BIS rates in `data/raw/BIS/mod.csv` by country/year to create `data/dta/fx.dta`. Rates are local currency per US dollar; the BIS source expresses euro-area legacy currencies in euros, including before adoption using fixed conversion rates. `do/data_cleaning/crosscountry_iv.do` uses the year-2000 sheets of the WIOD national tables and the cleaned commodity-price file `data/dta/comtrade_lit_energysafe_gassplice_cepiisplce11.dta`. It sums domestic and imported WIOD flows, divides by each fine using sector's intermediate inputs, and maps commodity prices through the supplied HS/WITS concordances. Prices are simple means within supplying sector/year; interior missing years are linearly interpolated. The log shock is $\log(\text{ind.tuv}/100)$. Matched IO shares are multiplied by the corresponding KLEMS year-2000 $\text{ii}/(\text{ii}+\text{comp})$; the multiplier is one when those data are missing. Unmatched supplying sectors are omitted without renormalizing the remaining shares. The FX variant adds $\log(\text{fx}/\text{fx.base})$ to each commodity shock, using 2000 FX or the first available year; missing FX adjustments are zero. Fine using-sector instruments are summed when mapped to grouped KLEMS industries. The output is `data/dta/crosscountry_iv.dta`; its unadjusted and FX-adjusted instruments are `shift_share_ct` and `shift_share_fx_ct`.

`do/data_cleaning/crosscountry_panel_finalize.do` merges the IV by country, industry and year, keeps manufacturing, and excludes Luxembourg and Slovakia, compensation above value added, intermediate inputs above gross output, $\text{ii}/\text{va}>10$, labor shares outside 0.05–0.95, and country/industry pairs with fewer than five years. Baseline intensity is the year-2000 ii/va , falling back to the first retained year. `treat=mat2va.base*lii_pi`; `va_base` is the sector's baseline value added divided by its country's total across retained sectors, using the same year-2000/first-year rule. Rows missing the outcome, treatment, FX instrument, wage or weight are excluded. The output `data/dta/crosscountry_panel.dta` contains 6,318 observations for 23 countries over 1995–2019, with key `country`, `indcode`, `year`; `ci` identifies country/industry pairs, and the FX instrument is renamed `shift_share_fx`. `do/tables/tab4.do` produces `output/tables/tab4.tex` from this panel.

`do/figures/fig_crosscountry.do` writes the four `figF1a.png`–`figF2b.png` panels to `output/figures/`. Figure F.2 chains changes in $\text{ii_pi}/\text{va_pi}$ within countries using current/lag intermediate expenditure shares. The European aggregation uses country intermediate-expenditure

totals in their retained source currency units; the dashed series subtracts log FX changes from changes in the relative-price index before aggregation.

3.4 Disaster instrument data (Tables D.1–D.3)

`do/data_cleaning/disaster_iv_construct.do` reads the archived EM-DAT workbook and the 1997 BACI trade/country CSVs in `data/raw/disaster_iv/`. It retains natural disasters starting in 1991–2016, excluding epidemics, infestations and animal incidents; missing death tolls become zero. A large event meets its country’s 90th percentile among positive-death events in that full window. The active country/year indicator covers the event year and the next two years, within the same window. BACI exports are summed from HS6 to HS4; a country/product is retained if it supplies at least 5% of world exports of that product and the product is at least 5% of the country’s exports. The HS4/year shock sums these exporters’ 1997 world shares times the active-disaster indicator.

The script maps HS4 shocks through `data/raw/Concordances/HS696_to_IOT.dta` using its supplied weights, then through `data/dta/IOLinkages.dta` and `data/raw/Concordances/IOT_NAICS.dta`. It converts NAICS 1997 to NAICS 2012 with `conc_naics97_naics12.dta` in the same concordance folder, averaging with annual shipments from `data/raw/NBER-CES/nberces_1958-2018.dta`. It merges onto the 1991–2016 NAICS/year skeleton from `data/dta/mainregfile.dta`; unmatched instrument values become zero. The output `data/dta/disaster_iv_data.dta` has key `naics`, `year` and contains the dimensionless weighted-exposure instrument `div_1`.

`do/tables/tabD1.do` separately constructs the disaster/product examples from these sources. `do/tables/tabD2.do` merges `div_1` into `mainregfile.dta` to estimate the disaster-IV specifications; `do/tables/tabD3.do` reports their actual first stages on the corresponding estimation samples. Their outputs are `output/tables/tabD1.tex`, `tabD2.tex` and `tabD3.tex` in the same folder.

`do/tables/tabD1.do` also writes `output/tables/tabD1_qualifying_countries.csv`, the full country list promised in the note to Table D.1. It contains one row for each of 201 countries with a qualifying disaster, before the exporter-specialization and top-20 filters: `iso3`, `Country`, `n_qualifying` (number of qualifying events), and `threshold` (the country-specific 90th-percentile death toll).

3.5 Quantitative-model variables and numerical construction

`do/data_cleaning/model_inputs.do` produces the two CSVs using the worksheet-level mapping in Table 2. Their identifiers are `year` (1998–2023), `sector_name` (the BEA category name) and `sector` (the integer encoding of the 22 names). The `sector-year` key is unique. Julia solves the total manufacturing series and each of the 19 nonaggregate subsectors separately; the durable, nondurable and total aggregates are excluded from the subsector loop.

In `quant_model/functions.jl`, `load_data` first restricts the data to years through `max_year`, then HP-filters each source series separately with smoothing parameter 5. The model constructs ratios *after* filtering the levels:

$$s_l = \text{Comp}/\text{GOmil}, \quad s_m = \text{Mat}/\text{GOmil}, \quad s_i = \text{Investmentmil}/\text{GOmil}, \quad \lambda = \text{Comp}/\text{VA}.$$

These are fractions, not percentages. The capital and output quantities are the BEA indexes `Cap_ind` and `GO_ind`; the investment price is `Investmentmil/Inv_ind`, in millions of dollars per quantity-index point. Wages are `Wage`, and materials prices are `Pm`. Expenditure `R`, passed to the model as `E`, is current-dollar `GOmil`; it is not deflated in this loader. An additional call with

smoothing zero supplies the unfiltered comparison series. Plot normalization is performed in `quant_model/plots.jl`.

`quant_model/main.jl` sets the substitution elasticity to 0.20, the adjustment-cost parameter to 0.75, the discount factor to $1/1.03$, and the solution horizon to 50 periods. The depreciation parameter is the mean of `Dep` across all 1998–2023 rows in the relevant input CSV, before applying the model’s 2019 input cutoff. The retained 1998–2019 input and 1998–2018 results windows are explained in the Julia software instructions below.

The main transition routines use outer tolerance 10^{-8} , maximum 20,000 iterations, damping 0.9 and investment-share damping 0.7. The material-share calculation in `functions.jl` uses an inner tolerance of 10^{-12} by default and 10^{-14} in the scalar steady-state calls. These tighter inner tolerances prevent numerical error in the nested calculation from stopping outer convergence. The verified run completed all 80 outer stages (four for manufacturing and four for each subsector) without changing published rounding.

The two counterfactuals, recovered fundamentals and manufacturing contribution statistics are calculated in `quant_model/main.jl` and serialized to `quant_model/results/results.jls`. The manufacturing contribution statistic uses sums of absolute deviations from the first observation, as in the paper’s formula. `quant_model/plots.jl` reads the cache to write the figures and compute Table E.1. That table reports average and endpoint changes in labor shares in percentage points: baseline changes are measured from the first baseline observation, and the price contributions are baseline minus counterfactual labor shares. Its value-added shares divide each subsector’s average unfiltered VA in 1998–2019 by the sum of these averages across the included subsectors and multiply by 100. Run `main.jl` first whenever inputs, calibration or model code change.

4 Software Requirements

Operating system and hardware

The complete replication was verified on September 10, 2026 under Windows 11 Enterprise LTSC (64-bit, build 26100), on an AMD Ryzen Threadripper PRO 7975WX (32 physical cores, 64 logical) with 256 GB of RAM. StataNow MP 19.5 was used. These are the tested machine’s specifications, not minimum hardware requirements. The Stata driver uses the package root selected as Stata’s working directory; Julia resolves its paths from the model directory. No high-performance computing, cluster or GPU was used. The largest single raw file read is about 517 MB.

Expected running time

Step	Measured time
<code>do/main.do</code> (all 53 Stata scripts)	16.43 minutes
<code>quant_model/main.jl</code> (model, figures, table)	79.00 seconds
Sum of Stata and Julia times	about 18 minutes

These timings are from the complete verification run on the machine described above. Generated datasets, model inputs, model cache and ordinary outputs were absent before execution. Dependencies had already been installed; installation on an empty machine was not part of this timing. Initial package downloads, building and precompilation require additional time that depends on the machine and network.

The unpacked package occupies approximately 38.0 GB (35.4 GiB), including 26.7 GB of raw inputs and 11.3 GB of non-proprietary CSV copies. Analysis datasets are rebuilt by the Stata

driver. Allow additional writable space for these datasets, temporary files, outputs and software dependencies; 38.0 GB describes the supplied files, not peak working storage during execution.

PyPlot renders the model figures through matplotlib. The verified Windows rendering environment uses Python 3.14.7 (conda-forge), matplotlib 3.11.1, NumPy 2.5.3, FreeType 2.14.3, and DejaVu Sans fonts (version 2.35). The Julia manifest does not pin these Python and font dependencies. Relative to the accepted figures, this environment produces small layout shifts (at most 1.13 points, or 0.40 mm); the plotted results agree at the paper’s reported precision. PDF metadata and rendering may differ across environments, so byte-identical PDFs are not expected.

Each Stata figure script calls `do/figures/graph_style.do` to select Stata’s built-in `stcolor` scheme and the Book Antiqua font. Both pixel width and height are specified for every exported panel, matching the accepted figures’ dimensions. Book Antiqua is the closest tested Windows match to the accepted figures’ Palatino-family appearance; the exact historical font configuration is unavailable. The font must be installed on the machine running Stata; font files are not bundled. If it is unavailable, Stata may substitute another font and alter label spacing. The scripts restore the caller’s previous scheme and font after successful execution or a Stata error. These settings apply to standalone figure runs as well as the master run. Small glyph and spacing differences can remain across font versions and operating systems.

Randomness

The empirical calculations and numerical model solution are deterministic. There is no random sampling, bootstrap or stochastic simulation, so no random seed needs to be set.

Stata — data construction and empirical results

Stata version 19.5 (StataNow, MP edition) was used. The Stata driver `do/main.do` checks for the following user-written packages and uses `ssc install` only when the corresponding command is missing:

```
labutil, coefplot, unique, gtools, egenmore, ereplace,  
reghdfe, ivreg2, ivreghdfe, estout, ranktest, ftools, binscatter,  
renvarlab
```

If missing, `binscatter2` is installed using `net install` from the fixed source revision <https://raw.githubusercontent.com/mdroste/stata-binscatter2/d4d5c523ff315c2c2cc6105d590894d09cb724e0>. The driver preserves existing installations, stops if an installation fails, and records the command locations and version headers in `replication_run.log`. Downloads require internet access. Missing SSC packages are obtained in the versions then available from SSC; their historical versions are not pinned or bundled. The September 10, 2026 verification used `reghdfe` 6.13.1, `ivreghdfe` 1.1.4, `ivreg2` 4.1.12, `ftools` 2.50.0, `gtools` 1.10.1, `estout` 3.33, `ranktest` 2.0.04, `coefplot` 1.8.8, `unique` 1.2.4, `ereplace` 1.0.3, `binscatter` 7.02, `binscatter2` 0.91, and `renvarlab` 1.0. These are the versions reported in the installed command files; the `labutil` and `egenmore` bundles do not expose a single package version in a matching command file.

The model-input builder `do/data_cleaning/model_inputs.do` uses only built-in Stata 19 commands; it requires no additional packages.

Stata temporary directory. The package-root path and Stata’s temporary-directory path may contain spaces. Stata’s temporary directory must be writable and have sufficient free space; its location is shown by `display c(tmpdir)`. On Windows, unpack into a short folder such as `C:`

/replication and use a short temporary-directory location so that full file paths stay below 260 characters. Deeply nested paths can cause Stata Excel exports to fail with `r(603)`, even when spaces are handled correctly.

Julia — quantitative model

Model data window and reported years. The model uses annual observations for 1998–2019; `max_year = 2019` sets the last input year. Recovering fundamentals requires one-period-ahead information, so reported model paths and counterfactual contribution statistics cover 1998–2018. The 2019 observation supplies the additional information needed to recover fundamentals through 2018. The HP filter is applied over the full 1998–2019 input window.

Julia 1.12.5. The packages loaded in `quant_model/loadpackages.jl` are `LinearAlgebra`, `Plots`, `CSV`, `DataFrames`, `StatsBase`, `LaTeXStrings`, `Parameters`, `Printf` and `Serialization`.

PyPlot is also required, and is the one dependency that is not loaded with `using`: `quant_model/plots.jl` selects the PyPlot backend by calling `pyplot()`, and `Plots` finds it as long as it is installed. PyPlot in turn needs a Python installation with `matplotlib`; PyCall bootstraps a private Conda one automatically if none is found.

`quant_model/loadpackages.jl` activates the supplied environment to install missing Julia packages with `Pkg.instantiate()`. The verified Julia package versions are recorded in `quant_model/Project.toml` and `quant_model/Manifest.toml`. Initial installation requires internet access. Python, `matplotlib` and fonts are separate dependencies, as described above; their versions are not pinned by the Julia manifest.

5 Instructions for Replicators

1. Download the replication package and keep the directory structure intact.
2. In Stata, set the working directory to the package root, the folder containing `do/`, `data/` and `quant_model/`:

```
cd "/path/to/Replication"
```

Replace the example path with the location of your copy. No code edits are required. The master initializes its root from this working directory, replacing any path left by an earlier run in the same Stata session.

3. Run the master in its entirety:

```
do "do/main.do"
```

It initializes paths, creates working directories and executes, in order: 21 data-cleaning scripts, then 13 figure scripts, then 19 table scripts. The first cleaning script, `do/data_cleaning/model_inputs.do`, rebuilds the two quantitative model input files in `quant_model/data/` from the archived BEA workbooks. Figure 2 is a schematic and does not require computational reproduction. It is supplied as `output/figures/fig2.png`. A full transcript of the Stata run is written to `replication_run.log` in the package root.

4. After constructing the model inputs in Stata, open a terminal in the `Replication` package root and run Julia 1.12.5:

```
julia --startup-file=no --project=quant_model quant_model/main.jl
```

Alternatively, run `quant_model/main.jl` from Julia. The script reads the rebuilt files from `quant_model/data/` and activates the supplied environment on startup. Missing Julia packages are installed automatically as described above.

To reproduce only the quantitative model, run these Stata commands:

```
cd "/path/to/Replication"  
do "do/setup.do"  
do "do/data_cleaning/model_inputs.do"
```

Then run `quant_model/main.jl` in Julia. Prebuilt input CSVs are included for convenience; the Stata step reproduces them from the raw workbooks without running the empirical analyses.

Running an individual Stata script. After the master has installed dependencies and constructed the analysis inputs, initialize the root before running a figure or table script. For example, to regenerate Table C.4:

```
cd "/path/to/Replication"  
do "do/setup.do"  
do "do/tables/tabC4.do"
```

The setup script validates the package location and initializes paths; it does not install dependencies, build inputs or create working directories. Individual scripts require this initialization and the same current working directory. If either is missing or inconsistent, they stop with an error. Run setup again after changing to another package copy. The model-only builder creates its own scratch and output directories and does not require a prior empirical run.

Rebuilding this documentation

The analysis exports table fragments and figure files; running it does not require a $\text{\LaTeX}$ installation. To rebuild this README, use a separate working directory containing `README.tex`, `data_citations.bib` and `ecta-fullname.bst`, and run:

```
pdflatex README.tex  
bibtex README  
pdflatex README.tex  
pdflatex README.tex
```

The result is `README.pdf`. The required $\text{\LaTeX}$ packages are listed in the source preamble.

6 Where Output Is Saved, and How It Maps to the Paper

Stata exhibits are written to `output/tables/` and `output/figures/`. Model exhibits are written to `quant_model/results/`. Output files are named after the exhibit number they produce, so `tabC6.tex` is Table C.6 and `figC3a.png` is the first panel of Figure C.3.

Numerical summaries outside the exhibit files. The Stata master records displayed regression output and diagnostics in `replication_run.log` at the package root. In particular, `do/figures/fig7.do` prints the two counterfactual contribution percentages under “Cumulative contribution” in that log. `quant_model/main.jl` prints the manufacturing contribution percentages for CF1 and CF2 in the Julia terminal and saves their unrounded values as `contribution_cf1` and `contribution_cf2` in `quant_model/results/results.jls`. This cache also contains the recovered fundamentals and baseline and counterfactual equilibrium paths. Julia terminal output is not automatically saved to a separate log. These locations supplement the figure and table crosswalk.

Tables

Exhibit	Script	Output file
Table 1	do/tables/tab1.do	output/tables/tab1.tex
Table 2	do/tables/tab2.do	output/tables/tab2.tex
Table 3	do/tables/tab3.do	output/tables/tab3.tex
Table 4	do/tables/tab4.do	output/tables/tab4.tex
Table C.1	do/tables/tabC1.do	output/tables/tabC1.tex
Table C.2	do/tables/tabC2.do	output/tables/tabC2.tex
Table C.3	do/tables/tabC3.do	output/tables/tabC3.tex
Table C.4	do/tables/tabC4.do	output/tables/tabC4.tex
Table C.5	do/tables/tabC5.do	output/tables/tabC5.tex
Table C.6	do/tables/tabC6.do	output/tables/tabC6.tex
Table C.7	do/tables/tabC7.do	output/tables/tabC7.tex
Table C.8	do/tables/tabC8.do	output/tables/tabC8.tex
Table C.9	do/tables/tabC9.do	output/tables/tabC9.tex
Table C.10	do/tables/tabC10.do	output/tables/tabC10.tex
Table C.11	do/tables/tabC11.do	output/tables/tabC11.tex
Table C.12	do/tables/tabC12.do	output/tables/tabC12.tex
Table D.1	do/tables/tabD1.do	output/tables/tabD1.tex
Table D.1 country list	do/tables/tabD1.do	output/tables/tabD1_qualifying_count ries.csv
Table D.2	do/tables/tabD2.do	output/tables/tabD2.tex
Table D.3	do/tables/tabD3.do	output/tables/tabD3.tex
Table B.1	—	Typeset directly in the paper; no computation
Table E.1	quant_model/main.jl	quant_model/results/subsector_contri butions_table.tex

Figures

Stata filenames below are relative to `output/figures/`; Julia filenames are relative to `quant_model/results/`. A range such as `a--c` means three separate files ending in `_a.pdf`, `_b.pdf` and `_c.pdf`.

Exhibit	Script	Output file(s)
Figure 1	do/figures/fig1.do	fig1.png
Figure 3	do/figures/fig3.do	fig3a.png, fig3b.png, fig3c.png
Figure 4	do/figures/fig4.do	fig4a.png, fig4b.png
Figure 5	do/figures/fig5.do	fig5a.png, fig5b.png, fig5a_ld.png, fig5b_ld.png
Figure 6	do/figures/fig6.do	fig6a.png, fig6b.png
Figure 7	do/figures/fig7.do	fig7a.png, fig7b.png
Figure B.1	do/figures/figB1.do	figB1.png
Figure C.1	do/figures/figC1.do	figC1.png

Exhibit	Script	Output file(s)
Figure C.2	do/figures/figC2.do	figC2.png
Figure C.3	do/figures/figC3.do	figC3a.png, figC3b.png
Figure C.4	do/figures/figC4.do	figC4.png
Figure C.5	do/figures/figC5.do	figC5.png
Figures F.1, F.2	do/figures/fig_crosscoun try.do	figF1a.png, figF1b.png, figF2a.png, figF2b.png
Figure 2	Supplied fixed schematic (no computation)	output/figures/fig2.png
Figure 8	quant_model/main.jl	main_inversion_a-c.pdf
Figure 9	quant_model/main.jl	main_counterfactuals_a-f.pdf
Figure E.1	quant_model/main.jl	appendix_inversion.pdf
Figure E.2	quant_model/main.jl	main_counterfactuals_g.pdf
Figure E.3	quant_model/main.jl	appendix_subsector_counterfactuals.p df

7 Data Citations

BANK FOR INTERNATIONAL SETTLEMENTS (2023): “US Dollar Exchange Rates,” Tech. rep., Bank for International Settlements, daily series.

BECKER, RANDY, WAYNE GRAY, AND JORDAN MARVAKOV (2016): “NBER-CES Manufacturing Industry Database: Technical Notes,” Tech. rep., National Bureau of Economic Research, 1958–2011 release.

——— (2021): “NBER-CES Manufacturing Industry Database: Technical Notes,” Tech. rep., National Bureau of Economic Research, 1958–2018 release.

BERTHOUE, ANTOINE AND CHARLOTTE EMLINGER (2011): “The Trade Unit Values Database,” Tech. rep., CEPII.

BONTADINI, FILIPPO, CAROL CORRADO, JONATHAN HASKEL, MASSIMILIANO IOMMI, AND CECILIA JONA-LASINIO (2025): “EU KLEMS & INTANProd Database,” Tech. rep., LUISS, 2025 release.

BUREAU OF ECONOMIC ANALYSIS (2003a): “Benchmark Input-Output Accounts: 1997 Use Table at NAICS Detail,” Tech. rep., U.S. Department of Commerce, accession NDN-0306.

——— (2003b): “Benchmark Input-Output Accounts: Concordance Between Input-Output Commodities and Foreign Trade Harmonized Codes,” Tech. rep., U.S. Department of Commerce, accession NDN-0317.

——— (2010): “GDP by Industry: SIC-Basis Historical Industry Accounts, 1947–1997,” Tech. rep., U.S. Department of Commerce.

——— (2019a): “GDP by Industry: Value Added and Gross Output by Industry,” Tech. rep., U.S. Department of Commerce, annual release of 29 October 2019.

——— (2019b): “National Income and Product Accounts, Section 6,” Tech. rep., U.S. Department of Commerce, release of 20 December 2019.

- (2022): “Regional Economic Accounts: CAGDP2 and CAINC30,” Tech. rep., U.S. Department of Commerce, vintages of December 2021 and June 2022.
- (2023): “BEA-BLS Integrated Industry-Level Production Account (KLEMS), 1987–2021,” Tech. rep., U.S. Department of Commerce.
- (2025a): “Fixed Assets Accounts, Section 3: Private Fixed Assets by Industry,” Tech. rep., U.S. Department of Commerce, tables 3.1ESI, 3.2ESI, 3.4ESI, 3.7ESI and 3.8ESI. Archived workbook records file creation on 15 September 2025; publication and original access dates not recorded.
- (2025b): “GDP by Industry: Gross Output, Intermediate Inputs and Value Added,” Tech. rep., U.S. Department of Commerce, archived workbooks identify publication on 25 September 2025; original access dates not recorded.
- (2025c): “National Income and Product Accounts, Section 6,” Tech. rep., U.S. Department of Commerce, tables 6.2D and 6.6D. Archived workbook identifies publication on 26 September 2025 and file creation on 22 December 2025; original access date not recorded.
- (2026a): “NAICS 2012 to Input-Output Industry Concordance,” Tech. rep., U.S. Department of Commerce.
- (2026b): “National Income and Product Accounts,” Tech. rep., U.S. Department of Commerce.
- BUREAU OF LABOR STATISTICS (2026): “Quarterly Census of Employment and Wages, Annual Averages, 1990–2020,” Tech. rep., U.S. Department of Labor.
- DELFORGE, DAMIEN, VALENTIN WATHELET, REGINA BELOW, CINZIA LANFREDI SOFIA, MARGO TONNELIER, JORIS A. F. VAN LOENHOUT, AND NIKO SPEYBROEK (2024): “The EM-DAT Emergency Events Database Archive,” Tech. rep., Open Data @ UCLouvain, version 2.0, archived April 30, 2026. <https://doi.org/10.14428/DVN/IOLTPH>. CC BY-NC-ND 4.0; accessed September 9, 2026 (UTC).
- DORN, DAVID (2009): “Essays on Inequality, Spatial Interaction, and the Demand for Skills,” Ph.D. thesis, University of St. Gallen, dissertation no. 3613.
- FALLY, THIBAUT AND JAMES SAYRE (2022): “Commodity Trade Matters,” Working Paper 24965, National Bureau of Economic Research.
- FEDERAL RESERVE BANK OF ST. LOUIS (2026): “FRED Economic Data,” Tech. rep., Federal Reserve Bank of St. Louis, 16 series.
- FERNALD, JOHN (2014): “A Quarterly, Utilization-Adjusted Series on Total Factor Productivity,” Tech. rep., Federal Reserve Bank of San Francisco, data vintage of March 2026.
- GAULIER, GUILLAUME AND SOLEDAD ZIGNAGO (2010): “BACI: International Trade Database at the Product-Level. The 1994–2007 Version,” Working Paper 2010-23, CEPII, data version V202501.
- SCHOTT, PETER K. (2019): “U.S. Imports and Exports by Detailed Product,” Tech. rep., Yale School of Management, legacy annual files; builds of July 2016 (imports) and December 2019 (exports).

- TIMMER, MARCEL P., ERIK DIETZENBACHER, BART LOS, ROBERT STEHRER, AND GAAITZEN J. DE VRIES (2015): “An Illustrated User Guide to the World Input-Output Database: The Case of Global Automotive Production,” *Review of International Economics*, 23, 575–605, wIOD 2016 release.
- UNITED NATIONS (2026): “UN Comtrade Database,” Tech. rep., United Nations Statistics Division.
- U.S. CENSUS BUREAU (2026a): “Economic Census, Concentration Ratios in Manufacturing,” Tech. rep., U.S. Department of Commerce, 1997, 2002, 2007 and 2012.
- (2026b): “North American Industry Classification System: Concordances and Industry Titles,” Tech. rep., U.S. Department of Commerce.
- U.S. ENERGY INFORMATION ADMINISTRATION (2026): “Natural Gas Imports Price,” Tech. rep., U.S. Department of Energy, series n9103us3A and n9100us3A.
- VIENNA INSTITUTE FOR INTERNATIONAL ECONOMIC STUDIES (2019): “EU KLEMS Growth and Productivity Accounts,” Tech. rep., wiiw, 2019 release.
- WORLD BANK (2025): “Commodity Price Data (The Pink Sheet),” Tech. rep., World Bank, updated January 2025.
- WORLD INTEGRATED TRADE SOLUTION (2026): “HS 1996 to ISIC Revision 3 Concordance,” Tech. rep., World Bank and UNCTAD.
- WORLD KLEMS (2015): “World KLEMS Accounts,” Tech. rep., World KLEMS, Canada, July 2012 release; Korea, 2015 file.